MODULE 3. LINEAR TRANSFORMATION

1. BASIC CONCEPT OF TRANSFORMATION

Let us think of a matrix equation $Ax = b$ in a different way. If we think of the matrix A as an object that acts on a vector $\mathbf{x}$ with respect to produce a new vector called Ax. For instance, the following equations define that multiplication by A transforms $\mathbf{x}$ into $\mathbf{b}$ and $\mathbf{u}$ into zero vector. From

$$\underset{\substack{\uparrow \\ A}}{\begin{bmatrix} 4 & -3 & 1 & 3 \\ 2 & 0 & 5 & 1 \end{bmatrix}} \underset{\substack{\uparrow \\ \mathbf{x}}}{\begin{bmatrix} 1 \\ 1 \\ 1 \\ 1 \end{bmatrix}} = \underset{\substack{\uparrow \\ \mathbf{b}}}{\begin{bmatrix} 5 \\ 8 \end{bmatrix}} \quad \text{and} \quad \underset{\substack{\uparrow \\ A}}{\begin{bmatrix} 4 & -3 & 1 & 3 \\ 2 & 0 & 5 & 1 \end{bmatrix}} \underset{\substack{\uparrow \\ \mathbf{u}}}{\begin{bmatrix} 1 \\ 4 \\ -1 \\ 3 \end{bmatrix}} = \underset{\substack{\uparrow \\ \mathbf{0}}}{\begin{bmatrix} 0 \\ 0 \end{bmatrix}}$$

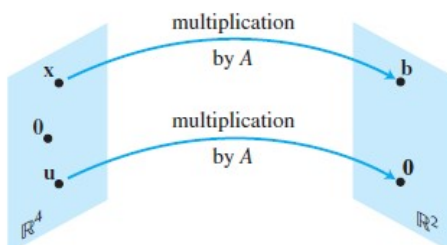


FIGURE 1 Transforming vectors via matrix multiplication.

this new point of view, solving the equation $A\mathbf{x} = \mathbf{b}$ indicates finding all vectors $\mathbf{x}$ in $\mathbb{R}^4$ that are transformed into the vector $\mathbf{b}$ in $\mathbb{R}^2$ under the action of multiplication by A.

Definition 1.1. ***Transformation:*** *A transformation (or function/mapping) T from $\mathbb{R}^n$ to $\mathbb{R}^m$, i.e., $T : \mathbb{R}^n \to \mathbb{R}^m$ is a rule that assigns a vector $T(\boldsymbol{x})$ in $\mathbb{R}^m$ to each vector $\boldsymbol{x}$ in $\mathbb{R}^n$. The set $\mathbb{R}^n$ is called the domain of T. For $\boldsymbol{x}$ in $\mathbb{R}^n$, the vector $T(\boldsymbol{x})$ in $\mathbb{R}^m$ is called the image of $\boldsymbol{x}$. The set of all images $T(\boldsymbol{x})$ is called the range of T.*

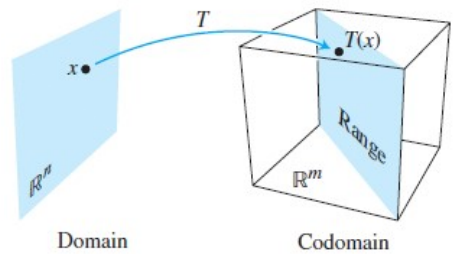


FIGURE 2 Domain, codomain, and range of $T : \mathbb{R}^n \to \mathbb{R}^m$.

2. MATRIX TRANSFORMATIONS

For each $\mathbf{x}$ in $\mathbb{R}^n$, $T(\mathbf{x})$ is computed as $A\mathbf{x}$, where A is $m \times n$ matrix. For simplicity, we sometimes denote such a matrix transformation by $\mathbf{x} \to A\mathbf{x}$.

Note:

1. The domain of T is $\mathbb{R}^n$ when A has n columns.
2. The codomain of T is $\mathbb{R}^m$ when each column of A has m entries.
3. The range of T is the set of all linear combinations of the columns of A because each image $T(\mathbf{x})$ is of the form $A\mathbf{x}$.

Example 1:

$$\text{Let } A = \begin{bmatrix} 1 & -3 \\ 3 & 5 \\ -1 & 7 \end{bmatrix}, \; u = \begin{bmatrix} 2 \\ -1 \end{bmatrix}, \; b = \begin{bmatrix} 3 \\ 2 \\ -5 \end{bmatrix}, \text{ and } c = \begin{bmatrix} 3 \\ 2 \\ 5 \end{bmatrix}.$$

Define a transformation $T : \mathbb{R}^2 \to \mathbb{R}^3$ by $T(\mathbf{x}) = A\mathbf{x}$ so that

$$T(x) = Ax = \begin{bmatrix} 1 & -3 \\ 3 & 5 \\ -1 & 7 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} = \begin{bmatrix} x_1 - 3x_2 \\ 3x_1 + 5x_2 \\ -x_1 + 7x_2 \end{bmatrix}.$$

a. Find $T(\mathbf{u})$, the image of u under T.

b. Find an $\mathbf{x}$ in $\mathbb{R}^2$ whose image under T is b.

c. Is there more than one $\mathbf{x}$ whose image under T is $\mathbf{b}$?

d. Determine if $\mathbf{c}$ is in the range of T.

Solution:

a)

$$T(u) = Au = \begin{bmatrix} 1 & -3 \\ 3 & 5 \\ -1 & 7 \end{bmatrix} \begin{bmatrix} 2 \\ -1 \end{bmatrix} = \begin{bmatrix} 5 \\ 1 \\ -9 \end{bmatrix}$$

b)

$$T(x) = b \Rightarrow Ax = b$$

$$\begin{bmatrix} 1 & -3 \\ 3 & 5 \\ -1 & 7 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} = \begin{bmatrix} 3 \\ 2 \\ -5 \end{bmatrix} \tag{2.1}$$

Applying row operations, we get

$$\left[\begin{array}{cc|c} 1 & -3 & 3 \\ 3 & 5 & 2 \\ -1 & 7 & -5 \end{array}\right] \backsim \left[\begin{array}{cc|c} 1 & -3 & 3 \\ 0 & 14 & -7 \\ 0 & 4 & -2 \end{array}\right] \backsim \left[\begin{array}{cc|c} 1 & -3 & 3 \\ 0 & 11 & -5 \\ 0 & 0 & 0 \end{array}\right] \backsim \left[\begin{array}{cc|c} 1 & 0 & \frac{3}{2} \\ 0 & 1 & -\frac{1}{2} \\ 0 & 0 & 0 \end{array}\right] \tag{2.2}$$

Hence, $x_1 = 1.5$ and $x_2 = -0.5$. $\therefore\ X = \begin{bmatrix} 1.5 \\ -0.5 \end{bmatrix}$.

c) From (2.2), it is clear that equation (2.1) has a unique solution. So, there is exactly one **x** whose image is **b**.

d) The vector **c** is in the range of T if **c** is the image of some **x** in $\mathbb{R}^2$, i.e., if $\mathbf{c} = T(\mathbf{x})$ for some **x**. $\therefore$ if $Ax = c$ is consistent, **c** will be in the range of T.

$$\left[\begin{array}{cc|c} 1 & -3 & 3 \\ 3 & 5 & 2 \\ -1 & 7 & 5 \end{array}\right] \backsim \left[\begin{array}{cc|c} 1 & -3 & 3 \\ 0 & 14 & -7 \\ 0 & 4 & 8 \end{array}\right] \backsim \left[\begin{array}{cc|c} 1 & -3 & 3 \\ 0 & 1 & 2 \\ 0 & 14 & -7 \end{array}\right] \backsim \left[\begin{array}{cc|c} 1 & -3 & 3 \\ 0 & 1 & 2 \\ 0 & 0 & -35 \end{array}\right]$$

Hence, the system is inconsistent and **c** is not in the range of T.

Example 2:

If $A = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 0 \end{bmatrix}$, then the transformation $\mathbf{x} \to A\mathbf{x}$ projects points in $\mathbb{R}^3$ onto

the x_1x_2-plane because $\begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} \to \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 0 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = \begin{bmatrix} x_1 \\ x_2 \\ 0 \end{bmatrix}$.

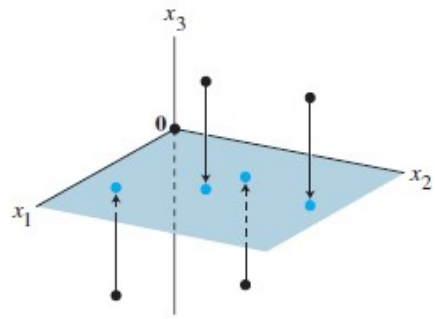


FIGURE 3
A projection transformation.

3. LINEAR TRANSFORMATIONS

Definition 3.1. ***Linear Transformation:*** *A transformation (or, mapping) T is linear if*

(1) $T(\mathbf{u}+\mathbf{v}) = T(\mathbf{u}) + T(\mathbf{v})$, $\forall$ $\mathbf{u},\ \mathbf{v}$ *in the domain of T;*
(2) $T(c\mathbf{u}) = cT(\mathbf{u})$, $\forall$ *scalars c and vectors $\mathbf{u}$ in the domain of T.*

Note:

1. Every matrix transformation is a linear transformation, not vice versa.
2. Linear transformation preserve the operations of vector addition and scalar multiplication.
3. If T is a linear transformation, then

$$T(0) = 0 \tag{3.1}$$

and

$$T(c\mathbf{u} + d\mathbf{v}) = cT(\mathbf{u}) + dT(\mathbf{v}), \tag{3.2}$$

$\forall$ vectors $\mathbf{u}, \mathbf{v}$ in the domain of T and all scalars $c,\ d$. The condition (3.2) is a combination of properties (1) and (2). If a transformation satisfies (3.2), it must be linear. If we set $c = d = 1$, we get preservation of addition and if we set $d = 0$, we get property (2). Repeated application of condition (3.2) produces a useful generalization;

$$T(c_1\mathbf{v_1} + \cdots + c_p\mathbf{v_p}) = c_1T(\mathbf{v_1}) + \cdots c_pT(\mathbf{v_p}). \tag{3.3}$$

Example 1: Given a scalar r, define $T : \mathbb{R}^2 \to \mathbb{R}^2$ by $T(\mathbf{x}) = r\mathbf{x}$. T is called a contraction when $0 \leq r \leq 1$ and a dilation when $r > 1$. Let $r = 3$ and show that T is a linear transformation.

Solution: Let **u, v** be in $\mathbb{R}^2$ and let $c,\ d$ be scalars. Then

$$T(c\mathbf{u}+d\mathbf{v}) = 3(c\mathbf{u}+d\mathbf{v}) = 3c\mathbf{u}+3d\mathbf{v} = c(3\mathbf{u})+d(3\mathbf{v}) = cT(\mathbf{u})+dT(\mathbf{v}).$$

Thus, T is a linear transformation because it satisfies property (3.2).

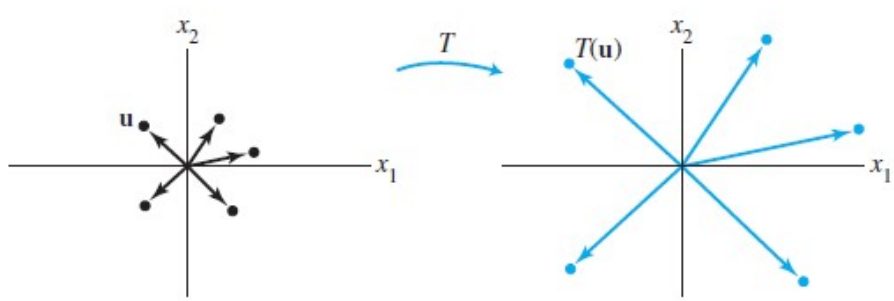


FIGURE 5 A dilation transformation.

Example 2: Define a linear transformation $T : \mathbb{R}^2 \to \mathbb{R}^2$ by

$$T(\mathbf{x}) = \begin{bmatrix} 0 & -1 \\ 1 & 0 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} = \begin{bmatrix} -x_2 \\ x_1 \end{bmatrix}$$

.

Find the images under T of $\mathbf{u} = \begin{pmatrix} 4 \\ 1 \end{pmatrix}$, $\mathbf{v} = \begin{pmatrix} 2 \\ 3 \end{pmatrix}$ and $\mathbf{u}+\mathbf{v} = \begin{pmatrix} 6 \\ 4 \end{pmatrix}$.

Solution:

$$T(\mathbf{u}) = \begin{bmatrix} 0 & -1 \\ 1 & 0 \end{bmatrix} \begin{pmatrix} 4 \\ 1 \end{pmatrix} = \begin{pmatrix} -1 \\ 4 \end{pmatrix},$$

$$T(\mathbf{v}) = \begin{bmatrix} 0 & -1 \\ 1 & 0 \end{bmatrix} \begin{pmatrix} 2 \\ 3 \end{pmatrix} = \begin{pmatrix} -3 \\ 2 \end{pmatrix},$$

$$T(\mathbf{u}+\mathbf{v}) = \begin{bmatrix} 0 & -1 \\ 1 & 0 \end{bmatrix} \begin{pmatrix} 6 \\ 4 \end{pmatrix} = \begin{pmatrix} -4 \\ 6 \end{pmatrix} = T(\mathbf{u}) + T(\mathbf{v}).$$

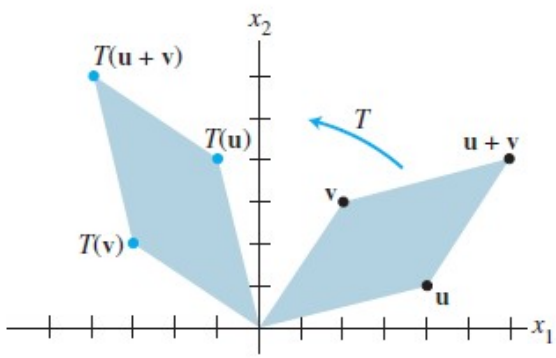


FIGURE 6 A rotation transformation.

Assignment

1. Let $A = \begin{pmatrix} 1/3 & 0 & 0 \\ 0 & 1/3 & 0 \\ 0 & 0 & 1/3 \end{pmatrix}$, $\mathbf{u} = \begin{pmatrix} 3 \\ 6 \\ -9 \end{pmatrix}$ and $\mathbf{v} = \begin{pmatrix} a \\ b \\ c \end{pmatrix}$. Define $T : \mathbb{R}^3 \to \mathbb{R}^3$ by $T(\mathbf{x}) = A\mathbf{x}$. Find $T(\mathbf{u})$ and $T(\mathbf{v})$.

2. Find a vector $\mathbf{x}$ whose image under T is $\mathbf{b}$, where $T(\mathbf{x}) = A\mathbf{x}$;

a) $A = \begin{pmatrix} 1 & 0 & -3 \\ -3 & 1 & 6 \\ 2 & -2 & -1 \end{pmatrix}$, $\mathbf{b} = \begin{pmatrix} -2 \\ 3 \\ -1 \end{pmatrix}$

b) $A = \begin{pmatrix} 1 & -2 & 3 \\ 0 & 1 & -3 \\ 2 & -5 & 6 \end{pmatrix}$, $\mathbf{b} = \begin{pmatrix} -6 \\ -4 \\ -5 \end{pmatrix}$

3. Let $A = \begin{pmatrix} 1 & -3 & 5 & -5 \\ 0 & 1 & -3 & 5 \\ 2 & -4 & 4 & -4 \end{pmatrix}$, $\mathbf{b} = \begin{pmatrix} -1 \\ 1 \\ 0 \end{pmatrix}$. Is $\mathbf{b}$ in the range of the linear transformation $\mathbf{x} \to A\mathbf{x}$?

4. Let $A = \begin{pmatrix} 3 & 2 & 10 & -6 \\ 1 & 0 & 2 & -4 \\ 0 & 1 & 2 & 3 \\ 1 & 4 & 10 & 8 \end{pmatrix}$, $\mathbf{b} = \begin{pmatrix} -1 \\ 3 \\ -1 \\ 4 \end{pmatrix}$. Is $\mathbf{b}$ in the range of the linear transformation $\mathbf{x} \to A\mathbf{x}$?

5. Define a linear transformation $T : \mathbb{R}^2 \to \mathbb{R}^2$ by

$$T(\mathbf{x}) = \begin{bmatrix} 2 & 0 \\ 0 & 2 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix}$$

.

Find the images under T of $\mathbf{u} = \begin{pmatrix} 5 \\ 2 \end{pmatrix}$, and $\mathbf{v} = \begin{pmatrix} -2 \\ 4 \end{pmatrix}$.

4. MATRIX OF LINEAR TRANSFORMATION

Example 1: The columns of $I_2 = \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix}$ are $e_1 = \begin{pmatrix} 1 \\ 0 \end{pmatrix}$ and $e_2 = \begin{pmatrix} 0 \\ 1 \end{pmatrix}$. Suppose T is a linear transformation from $\mathbb{R}^2$ into $\mathbb{R}^3$ such that $T(e_1) = \begin{pmatrix} 5 \\ -7 \\ 2 \end{pmatrix}$ and $T(e_2) = \begin{pmatrix} -3 \\ 8 \\ 0 \end{pmatrix}$. Find a formula for the image of an arbitrary $\mathbf{x}$ in $\mathbb{R}^2$.

Solution: Let $\mathbf{x} = \begin{pmatrix} x_1 \\ x_2 \end{pmatrix} = x_1 \begin{pmatrix} 1 \\ 0 \end{pmatrix} + x_2 \begin{pmatrix} 0 \\ 1 \end{pmatrix} = x_1 e_1 + x_2 e_2$.

$\because T$ is a linear transformation,

$$T(x) = x_1 T(e_1) + x_2 T(e_2) = x_1 \begin{pmatrix} 5 \\ -7 \\ 2 \end{pmatrix} + x_2 \begin{pmatrix} -3 \\ 8 \\ 0 \end{pmatrix} = \begin{pmatrix} 5x_1 - 3x_2 \\ -7x_1 + 8x_2 \\ 2x_1 \end{pmatrix}$$

$\therefore T(e_1)$ and $T(e_2)$ are sufficient to determine $T(x)$ for any x.

$T(x) = x_1T(e_1) + x_2T(e_2)$ can be rewritten as

$$T(x) = (T(e_1)T(e_2)) \begin{pmatrix} x_1 \\ x_2 \end{pmatrix} = Ax.$$

Theorem: Let $T : \mathbb{R}^n \to \mathbb{R}^m$ be a linear transformation. Then there exists a unique matrix A such that

$$T(\mathbf{x}) = A\mathbf{x},\ \forall\, \mathbf{x} \text{ in } \mathbb{R}^n,\ \text{where } A = (T(e_1)\ T(e_2) \cdots T(e_n))\,.$$

The matrix A, known as the **standard matrix for the linear transformation** T, is a $m \times n$ matrix whose jth column is the vector $T(e_j)$, where e_j is the jth column of the identity matrix in $\mathbb{R}^n$.

Note: Every linear transformation from $\mathbb{R}^n$ to $\mathbb{R}^m$ can be viewed as a matrix transformation and vice versa. The term linear transformation focuses on a property of a mapping, while matrix transformation describes how such a mapping is implemented.

Example 2: Find the standard matrix A for the dilation transformation $T(\mathbf{x}) = 3\mathbf{x}$ for $\mathbf{x}$ in $\mathbb{R}^2$.

Solution: Let us consider $e_1 = \begin{pmatrix} 1 \\ 0 \end{pmatrix}$ and $e_2 = \begin{pmatrix} 0 \\ 1 \end{pmatrix}$.

So, $T(e_1) = 3e_1 = \begin{pmatrix} 3 \\ 0 \end{pmatrix}$, $T(e_2) = 3e_2 = \begin{pmatrix} 0 \\ 3 \end{pmatrix}$, $\therefore\ A = \begin{pmatrix} 3 & 0 \\ 0 & 3 \end{pmatrix}$

TABLE 1 Reflections

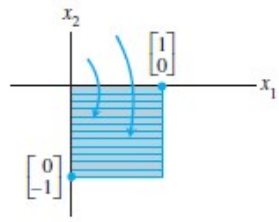
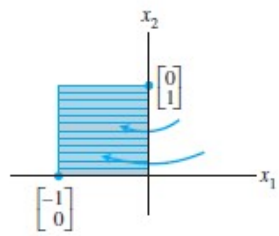
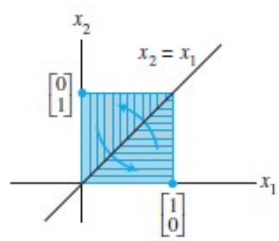
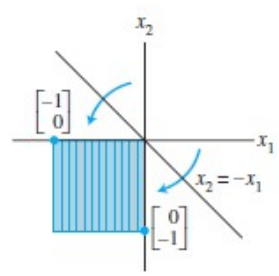
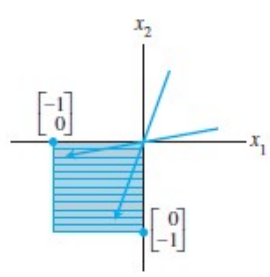
Transformation	Image of the Unit Square	Standard Matrix
Reflection through the x_1-axis		$\begin{bmatrix}1 & 0\\0 & -1\end{bmatrix}$
Reflection through the x_2-axis		$\begin{bmatrix}-1 & 0\\0 & 1\end{bmatrix}$
Reflection through the line $x_2 = x_1$		$\begin{bmatrix}0 & 1\\1 & 0\end{bmatrix}$
Reflection through the line $x_2 = -x_1$		$\begin{bmatrix}0 & -1\\-1 & 0\end{bmatrix}$
Reflection through the origin		$\begin{bmatrix}-1 & 0\\0 & -1\end{bmatrix}$

TABLE 2 Contractions and Expansions

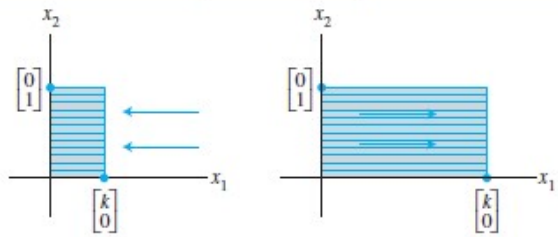
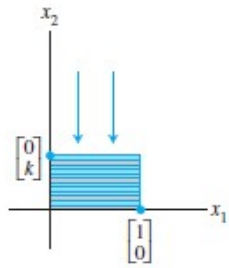
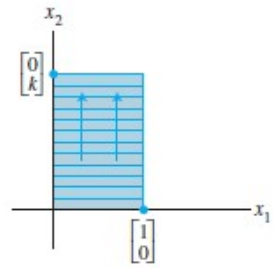
Transformation	Image of the Unit Square	Standard Matrix
Horizontal contraction and expansion	 $0 < k < 1$ $k > 1$	$\begin{bmatrix}k & 0\\0 & 1\end{bmatrix}$
Vertical contraction and expansion	 $0 < k < 1$  $k > 1$	$\begin{bmatrix}1 & 0\\0 & k\end{bmatrix}$

TABLE 3 Shears

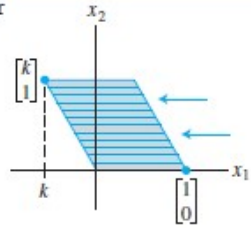
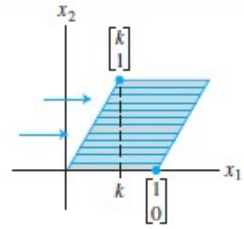
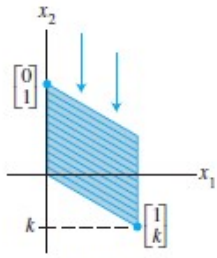
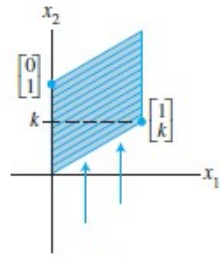
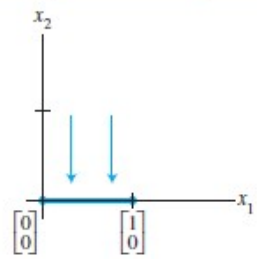
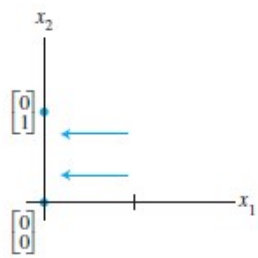
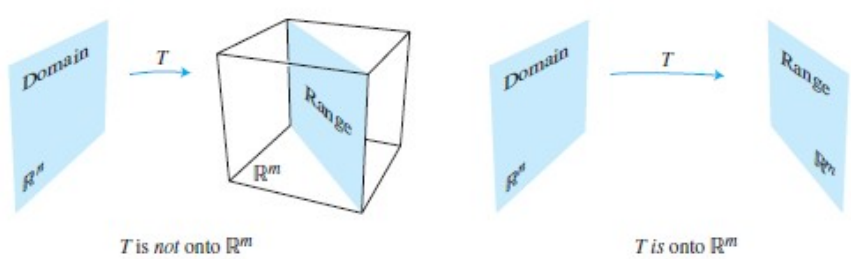
Transformation	Image of the Unit Square	Standard Matrix
Horizontal shear	 $k < 0$  $k > 0$	$\begin{bmatrix}1 & k\\0 & 1\end{bmatrix}$
Vertical shear	 $k < 0$  $k > 0$	$\begin{bmatrix}1 & 0\\k & 1\end{bmatrix}$

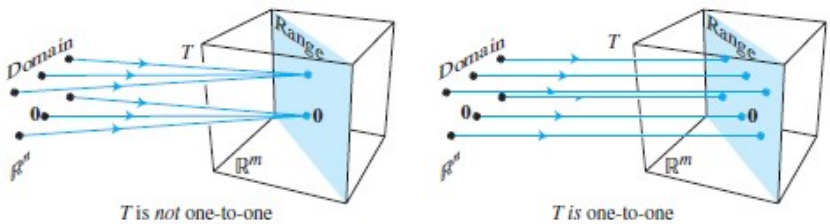
TABLE 4 Projections

Transformation	Image of the Unit Square	Standard Matrix
Projection onto the x_1-axis		$\begin{bmatrix}1 & 0\\0 & 0\end{bmatrix}$
Projection onto the x_2-axis		$\begin{bmatrix}0 & 0\\0 & 1\end{bmatrix}$

Definition 4.1. ***Onto mapping:*** *A mapping $T : \mathbb{R}^n \to \mathbb{R}^m$ is said to be onto $\mathbb{R}^m$ if each $\mathbf{b}$ in $\mathbb{R}^m$ is the image of atleast one x in $\mathbb{R}^n$, i.e., for each $\mathbf{b}$ in $\mathbb{R}^m$, $\exists$ atleast one solution of $T(\mathbf{x}) = \mathbf{b}$.*



Definition 4.2. ***One-to-one mapping:*** *A mapping $T : \mathbb{R}^n \to \mathbb{R}^m$ is said to be one-to-one if each $\mathbf{b}$ in $\mathbb{R}^m$ is the image of atmost one $\mathbf{x}$ in $\mathbb{R}^n$, i.e., for each $\mathbf{b}$ in $\mathbb{R}^m$, the equation $T(\mathbf{x}) = \mathbf{b}$ has either a unique solution or none at all.*



Theorem: Let A be an $m \times n$ matrix. Then the following statements are logically equivalent, i.e., for a particular A, either they are all true statements or they are all false;

a) For each $\mathbf{b}$ in $\mathbb{R}^m$, the equation $A\mathbf{x} = \mathbf{b}$ has a solution.

b) Each $\mathbf{b}$ in $\mathbb{R}^m$ is a linear combination of the columns of A.

c) The columns of A span $\mathbb{R}^m$.

d) A has a pivot position in every row.

Example 3: Let T be the linear transformation whose standard matrix is

$$A = \begin{bmatrix} 1 & -4 & 8 & 1 \\ 0 & 2 & -1 & 3 \\ 0 & 0 & 0 & 5 \end{bmatrix}$$

Does T map $\mathbb{R}^4$ onto $\mathbb{R}^3$? Is T a one-to-one mapping?

Solution: $\because$ A is in echelon form, we can see that A has a pivot position in each row. From the above theorem, we can say that for each $\mathbf{b}$ in $\mathbb{R}^3$, $A\mathbf{x} = \mathbf{b}$ is consistent. In other words, T maps $\mathbb{R}^4$ onto $\mathbb{R}^3$. However, $\because$ the equation $A\mathbf{x} = \mathbf{b}$ has a free variable, each $\mathbf{b}$ is in the image of more than one $\mathbf{x}$, i.e., T is not one-to-one.

Theorem:

1. Let $T : \mathbb{R}^n \to \mathbb{R}^m$ be a linear transformation. Then T is one-to-one iff $T(\mathbf{x}) = 0$ has only the trivial solution.
2. Let $T : \mathbb{R}^n \to \mathbb{R}^m$ be a linear transformation and let A be the standard matrix for T. Then

a) T maps $\mathbb{R}^n$ onto $\mathbb{R}^m$ iff the columns of A span $\mathbb{R}^m$.

b) T is one-to-one iff the columns of A are linearly independent.

Example 4: Let $T(x_1, x_2) = (3x_1 + x_2, 5x_1 + 7x_2, x_1 + 3x_2)$. Show that T is a one-to-one linear transformation. Does T map $\mathbb{R}^2$ onto $\mathbb{R}^3$?

Solution:

$$T(x) = \begin{pmatrix} 3x_1 + x_2 \\ 5x_1 + 7x_2 \\ x_1 + 3x_2 \end{pmatrix} = \begin{pmatrix} 3 & 1 \\ 5 & 7 \\ 1 & 3 \end{pmatrix} \begin{pmatrix} x_1 \\ x_2 \end{pmatrix}$$

$\therefore T$ is linear transformation.

The columns of A are linearly independent because they are not multiples. By above theorem,T is one-to-one.

$\because A$ is 3×2 matrix, the columns of A span $\mathbb{R}^3$ iff A has 3 pivot positions by theorem 2.b). This is impossible since A has only 2 columns. So the columns of A do not span $\mathbb{R}^3$ and the linear transformation is not onto $\mathbb{R}^3$.

Assignment

1. Find the standard matrix of T;

a) $T : \mathbb{R}^2 \to \mathbb{R}^4$ by $T(\mathbf{e_1}) = (3, 1, 3, 1)$ and $T(\mathbf{e_2}) = (-5, 2, 0, 0)$ where $\mathbf{e_1} = (1, 0)$ and $\mathbf{e_2} = (0, 1)$.

b) $T : \mathbb{R}^3 \to \mathbb{R}^2$ by $T(\mathbf{e_1}) = (1, 4)$, $T(\mathbf{e_2}) = (-2, 9)$, and $T(\mathbf{e_3}) = (3, -8)$ where $\mathbf{e_1} = (1, 0, 0)$, $\mathbf{e_2} = (0, 1, 0)$, and $\mathbf{e_3} = (0, 0, 1)$.

c) $T : \mathbb{R}^2 \to \mathbb{R}^2$ is a vertical shear transformation that maps $\mathbf{e_1}$ into $\mathbf{e_1} - 3\mathbf{e_2}$ but leaves $\mathbf{e_2}$ unchanged.

2. Let $T : \mathbb{R}^2 \to \mathbb{R}^2$ be a linear transformation such that $T(x_1, x_2) = (x_1 + x_2, 4x_1 + 5x_2)$. Find $\mathbf{x}$ such that $T(\mathbf{x}) = (3, 8)$.

3. Find the standard matrix for the transformation $T \begin{pmatrix} x_1 \\ x_2 \\ x_3 \end{pmatrix} = \begin{pmatrix} 2x_1 - 4x_2 \\ x_1 - x_3 \\ -x_2 + 3x_3 \end{pmatrix}$.

4. Let T be the linear transformation whose standard matrix is

$$A = \begin{bmatrix} -5 & 6 & -5 & -6 \\ 8 & 3 & -3 & 8 \\ 2 & 9 & 5 & -12 \\ -3 & 2 & 7 & -12 \end{bmatrix}$$

Is T a one-to-one mapping?

5. Show that the transformation T defined by $T(x_1, x_2) = (x_1 - 2x_2, x_1 - 3, 2x_1 - 5x_2)$ is not linear.

5. INNER PRODUCT

Definition 5.1. ***Inner product:*** *If* $\boldsymbol{u}$ *and* $\boldsymbol{v}$ *are vectors in* $\mathbb{R}^n$*, then we regard* $\boldsymbol{u}$ *and* $\boldsymbol{v}$ *as* $n \times 1$ *matrices. The transpose* $\boldsymbol{u}^T$ *is a* $1 \times n$ *matrix, and the matrix product* $\boldsymbol{u}^T\boldsymbol{v}$ *is a* 1×1 *matrix, which we write as a single real number (a scalar) without bracket. The number* $\boldsymbol{u}^T\boldsymbol{v}$ *is called the* ***inner product*** *of* $\boldsymbol{u}$ *and* $\boldsymbol{v}$*, and often it is written as* $\boldsymbol{u}\cdot\boldsymbol{v}$*. This inner product is also referred to as a dot product. If*

$$\mathbf{u} = \begin{bmatrix} u_1 \\ u_2 \\ \vdots \\ u_n \end{bmatrix} \text{ and } \mathbf{v} = \begin{bmatrix} v_1 \\ v_2 \\ \vdots \\ v_n \end{bmatrix}$$

then the inner product of $\boldsymbol{u}$ *and* $\boldsymbol{v}$ *is*

$$[u_1\ u_2 \cdots\ u_n] \begin{bmatrix} v_1 \\ v_2 \\ \vdots \\ v_n \end{bmatrix} = u_1v_1 + u_2v_2 + \cdots + u_nv_n.$$

Example 1: Compute $\mathbf{u}\cdot\mathbf{v}$ and $\mathbf{v}\cdot\mathbf{u}$ for $\mathbf{u} = \begin{bmatrix} 2 \\ -5 \\ -1 \end{bmatrix}$ and $\mathbf{v} = \begin{bmatrix} 3 \\ 2 \\ -3 \end{bmatrix}$.

Solution: $\mathbf{u}\cdot\mathbf{v} = \mathbf{u}^T\mathbf{v} = \begin{bmatrix} 2 & -5 & -1 \end{bmatrix} \begin{bmatrix} 3 \\ 2 \\ -3 \end{bmatrix} = 2 \cdot 3 + (-5) \cdot 2 + (-1)(-3) = -1$

$\mathbf{v}\cdot\mathbf{u} = \mathbf{v}^T\mathbf{u} = \begin{bmatrix} 3 & 2 & -3 \end{bmatrix} \begin{bmatrix} 2 \\ -5 \\ -1 \end{bmatrix} = 3 \cdot 2 + (2)(-5) + (-3)(-1) = -1.$

Theorem: Let $\mathbf{u}$, $\mathbf{v}$, and $\mathbf{w}$ be vectors in $\mathbb{R}^n$ and c be a scalar. Then

a) $\mathbf{u}\cdot\mathbf{v} = \mathbf{v}\cdot\mathbf{u}$

b) $(\mathbf{u} + \mathbf{v}) \cdot \mathbf{w} = \mathbf{u} \cdot \mathbf{w} + \mathbf{v} \cdot \mathbf{w}$

c) $(c\mathbf{u}) \cdot \mathbf{v} = c(\mathbf{u}\cdot\mathbf{v}) = \mathbf{u} \cdot (c\mathbf{v})$

d) $\mathbf{u}\cdot\mathbf{u} \geq 0$ and $\mathbf{u}\cdot\mathbf{u} = 0$ iff $\mathbf{u} = 0$.

Properties b) and c) can be combined several times to produce the useful rule given by

$$(c_1\mathbf{u}_1 + \cdots + c_p\mathbf{u}_p)\cdot\mathbf{w} = c_1(\mathbf{u}_1\cdot\mathbf{w}) + \cdots + c_p(\mathbf{u}_p\cdot\mathbf{w}).$$

Definition 5.2. ***Length of a vector:*** *If* $\boldsymbol{v}$ *is in* $\mathbb{R}^n$*, with entries* $v_1,\ v_2, \cdots,\ v_n$*, then the square root of* $\boldsymbol{v}\cdot\boldsymbol{v}$ *is defined because* $\boldsymbol{v}\cdot\boldsymbol{v}$ *is non-negative. The* ***length*** *(or* ***norm****) of* $\boldsymbol{v}$ *is the non-negative scalar* $\| \boldsymbol{v} \|$ *defined by*

$$\| \boldsymbol{v} \| = \sqrt{\boldsymbol{v}\cdot\boldsymbol{v}} = \sqrt{v_1^2 + v_2^2 + \cdots + v_n^2},\ \textit{and}\ \| \boldsymbol{v} \|^2 = \boldsymbol{v}\cdot\boldsymbol{v}.$$

Note:

1. For any scalar c, the length of $c\mathbf{v}$ is $| c |$ times the length of $\mathbf{v}$, i.e., $\| c\mathbf{v} \| = | c | \| \mathbf{v} \|$.
   (Compute $\| c\mathbf{v} \|^2 = (c\mathbf{v}) \cdot (c\mathbf{v}) = c^2\mathbf{v} \cdot \mathbf{v} = c^2 \| \mathbf{v} \|^2$ and take square roots)
2. A vector whose length is 1 is called a unit vector. If we divide a non-zero vector $\mathbf{v}$ by its length (i.e. $\mathbf{v}/ \| \mathbf{v} \|$), we obtain a unit vector $\mathbf{u}$.

The process of creating **u** from **v** is sometimes called normalizing **v** and we say that **u** is in the same direction as **v**.

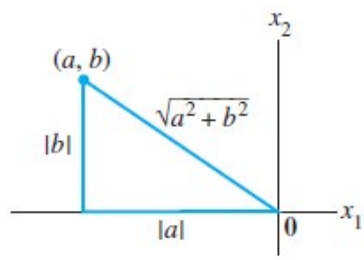


FIGURE 1. Interpretation of $\| \mathbf{v} \|$ as length

Example 2: Let $\mathbf{v} = (1, -2, 2, 0)$. Find a unit vector **u** in the same direction as **v**.
Solution: $\| \mathbf{v} \|^2 = \mathbf{v} \cdot \mathbf{v} = (1)^2 + (-2)^2 + (2)^2 + (0)^2 = 9, \therefore \| \mathbf{v} \| = 3$.
Now, $\mathbf{u} = \frac{\mathbf{v}}{\|\mathbf{v}\|} = (\frac{1}{3}, -\frac{2}{3}, \frac{2}{3}, 0)$.

Example 3: Let W be the subspace spanned by $\mathbf{x} = (2/3, 1)$. Find a unit vector z that is a basis for W.
Solution: $\because W$ consists of all multiples of x, any non-zero vector in W is a basis for W. Now, to simplify the calculation, $\mathbf{y} = 3\mathbf{x} = (2, 3)$.
Computing $\| \mathbf{y} \|^2 = 13$ and normalizing **y**, we get $\mathbf{z} = (2/\sqrt{13}, 3/\sqrt{13})$.
Hence, another basis for W is $(2/\sqrt{13}, 3/\sqrt{13})$.

Definition 5.3. ***Distance between two vectors:*** *For* $\boldsymbol{u}$ *and* $\boldsymbol{v}$ *in* $\mathbb{R}^n$*, the* ***distance between*** $\boldsymbol{u}$ *and* $\boldsymbol{v}$*, written as dist*$(\boldsymbol{u}, \boldsymbol{v})$*, is the length of the vector* $\boldsymbol{u} - \boldsymbol{v}$*, i.e., dist*$(\boldsymbol{u}, \boldsymbol{v}) = \| \boldsymbol{u} - \boldsymbol{v} \|$.

Example 4: Compute the distance between the vectors $\mathbf{u} = (7, 1)$ and $\mathbf{v} = (3, 2)$.
Solution: $\mathbf{u} - \mathbf{v} = (7, 1) - (3, 2) = (4, -1)$
$\therefore \| \mathbf{u} - \mathbf{v} \| = \sqrt{4^2 + (-1)^2} = \sqrt{17}$.

6. ORTHOGONALITY

Definition 6.1. ***Orthogonal vectors:*** *Two vectors* $\boldsymbol{u}$ *and* $\boldsymbol{v}$ *in* $\mathbb{R}^n$ *are* ***orthogonal*** *(to each other) if* $\boldsymbol{u} \cdot \boldsymbol{v} = 0$.

Note: The zero vector is orthogonal to every vector in $\mathbb{R}^n$ because $\mathbf{0}^T\mathbf{v} = 0$ for all $\mathbf{v}$.

The Pythagorean theorem: Two vectors $\mathbf{u}$ and $\mathbf{v}$ are orthogonal iff

$$\| \mathbf{u} + \mathbf{v} \|^2 = \| \mathbf{u} \|^2 + \| \mathbf{v} \|^2 .$$

Definition 6.2. ***Orthogonal complements:*** *If a vector* $\boldsymbol{z}$ *is orthogonal to every vector in a subspace* W *of* $\mathbb{R}^n$*, then* $\boldsymbol{z}$ *is said to be orthogonal to* W*. The set of all vectors* $\boldsymbol{z}$ *that are orthogonal to* W *is called the* ***orthogonal complement*** *of* W *and is denoted by* $W^\perp$ *(and read as '*W *perpendicular').*

Note:

1. A vector $\mathbf{x}$ is in $W^\perp$ iff $\mathbf{x}$ is orthogonal to every vector in a set that spans W.
2. $W^\perp$ is a subspace of $\mathbb{R}^n$.

Example 1:

Let $\mathbf{a} = \begin{bmatrix} -2 \\ 1 \end{bmatrix}$, $\mathbf{b} = \begin{bmatrix} -3 \\ 1 \end{bmatrix}$, $\mathbf{c} = \begin{bmatrix} 4/3 \\ -1 \\ 2/3 \end{bmatrix}$, and $\mathbf{d} = \begin{bmatrix} 5 \\ 6 \\ -1 \end{bmatrix}$.

a) Compute $\frac{\mathbf{a}\cdot\mathbf{b}}{\mathbf{a}\cdot\mathbf{a}}$ and $\left(\frac{\mathbf{a}\cdot\mathbf{b}}{\mathbf{a}\cdot\mathbf{a}}\right)\mathbf{a}$.

b) Find a unit vector $\mathbf{u}$ in the direction of $\mathbf{c}$.

c) Show that $\mathbf{d}$ is orthogonal to $\mathbf{c}$.

Solution: a) $\mathbf{a}\cdot\mathbf{b} = 7$, $\mathbf{a}\cdot\mathbf{a} = 5$. Hence, $\frac{\mathbf{a}\cdot\mathbf{b}}{\mathbf{a}\cdot\mathbf{a}} = \frac{7}{5}$ and $\left(\frac{\mathbf{a}\cdot\mathbf{b}}{\mathbf{a}\cdot\mathbf{a}}\right)\mathbf{a} = \begin{bmatrix} -14/5 \\ 7/5 \end{bmatrix}$.

b) $\| \mathbf{c} \| = \sqrt{29}/3$. Hence, $\mathbf{u} = \frac{\mathbf{c}}{\|\mathbf{c}\|} = \begin{bmatrix} 4/\sqrt{29} \\ -3/\sqrt{29} \\ 2/\sqrt{29} \end{bmatrix}$.

c) $\mathbf{d}$ is orthogonal to $\mathbf{c}$ because $\mathbf{d} \cdot \mathbf{c} = \begin{bmatrix} 5 \\ 6 \\ -1 \end{bmatrix} \cdot \begin{bmatrix} 4/3 \\ -1 \\ 2/3 \end{bmatrix} = 0$.

Assignment

1. Compute the following quantities using the vectors $\mathbf{u} = \begin{bmatrix} -1 \\ 2 \end{bmatrix}$, $\mathbf{v} = \begin{bmatrix} 4 \\ 6 \end{bmatrix}$, $\mathbf{w} = \begin{bmatrix} 3 \\ -1 \\ -5 \end{bmatrix}$, and $\mathbf{x} = \begin{bmatrix} 6 \\ -2 \\ 3 \end{bmatrix}$;

a) $\mathbf{u} \cdot \mathbf{u}$, $\mathbf{v} \cdot \mathbf{u}$, and $\frac{\mathbf{v} \cdot \mathbf{u}}{\mathbf{u} \cdot \mathbf{u}}$, b) $\frac{\mathbf{u}}{\mathbf{u} \cdot \mathbf{u}}$, c) $\frac{\mathbf{w}}{\mathbf{w} \cdot \mathbf{w}}$, d) $\| \mathbf{x} \|$, e) $\left(\frac{\mathbf{x} \cdot \mathbf{w}}{\mathbf{x} \cdot \mathbf{x}}\right) \mathbf{x}$.

2. Find unit vectors in the direction of the given vectors;

a) $\begin{bmatrix} -30 \\ 40 \end{bmatrix}$, b) $\begin{bmatrix} 8/3 \\ 2 \end{bmatrix}$, c) $\begin{bmatrix} 7/4 \\ 1/2 \\ 1 \end{bmatrix}$.

3. Find the distance between $\mathbf{x} = \begin{bmatrix} 10 \\ -3 \end{bmatrix}$ and $\mathbf{y} = \begin{bmatrix} -1 \\ -5 \end{bmatrix}$.

4. Determine which pairs of vectors are orthogonal;

a) $\mathbf{a} = \begin{bmatrix} 8 \\ -5 \end{bmatrix}$ and $\mathbf{b} = \begin{bmatrix} -2 \\ -3 \end{bmatrix}$

b) $\mathbf{u} = \begin{bmatrix} 12 \\ 3 \\ -5 \end{bmatrix}$ and $\mathbf{v} = \begin{bmatrix} 2 \\ -3 \\ 3 \end{bmatrix}$

c) $\mathbf{y} = \begin{bmatrix} -3 \\ 7 \\ 4 \\ 0 \end{bmatrix}$ and $\mathbf{z} = \begin{bmatrix} 1 \\ -8 \\ 15 \\ -7 \end{bmatrix}$

Definition 6.3. ***Orthogonal sets:*** *A set of vectors $\{\boldsymbol{u}_1, \cdots, \boldsymbol{u}_p\}$ in $\mathbb{R}^n$ is said to be* ***orthogonal set*** *if each pair of distinct vectors from the set is orthogonal, i.e., if $\boldsymbol{u}_i \cdot \boldsymbol{u}_j = 0$ whenever $i \neq j$.*

Example 2: Show that $\{\mathbf{u}_1, \mathbf{u}_2, \mathbf{u}_3\}$ is an orthogonal set, where

$$\mathbf{u}_1 = \begin{bmatrix} 3 \\ 1 \\ 1 \end{bmatrix}, \mathbf{u}_2 = \begin{bmatrix} -1 \\ 2 \\ 1 \end{bmatrix}, \text{ and } \mathbf{u}_3 = \begin{bmatrix} -1/2 \\ -2 \\ 7/2 \end{bmatrix}.$$

Solution: Let us consider the three possible pairs of distinct vectors, namely $\{\mathbf{u}_1, \mathbf{u}_2\}$, $\{\mathbf{u}_1, \mathbf{u}_3\}$, and $\{\mathbf{u}_2, \mathbf{u}_3\}$.

$$\mathbf{u}_1 \cdot \mathbf{u}_2 = 3(-1) + 1(2) + 1(1) = 0$$

$$\mathbf{u}_1 \cdot \mathbf{u}_3 = 3(-1/2) + 1(-2) + 1(7/2) = 0$$

$$\mathbf{u}_2 \cdot \mathbf{u}_3 = -1(-1/2) + 2(-2) + 1(7/2) = 0$$

Each pair of distinct vectors is orthogonal and so $\{\mathbf{u}_1, \mathbf{u}_2, \mathbf{u}_3\}$ is an orthogonal set.

Theorem: If $S = \{\mathbf{u}_1, \cdots, \mathbf{u}_p\}$ is an orthogonal set of non-zero vectors in $\mathbb{R}^n$, then S is linearly independent and hence is a basis for the subspace spanned by S.

Definition 6.4. ***Orthogonal basis:*** *An* ***orthogonal basis*** *for a subspace W of $\mathbb{R}^n$ is a basis for W that is also an orthogonal set.*

Theorem: Let $\{\mathbf{u}_1, \cdots, \mathbf{u}_p\}$ be an orthogonal basis for a subspace W of $\mathbb{R}^n$. For each $\mathbf{y}$ in W, the weights in the linear combination

$$\mathbf{y} = c_1\mathbf{u}_1 + \cdots + c_p\mathbf{u}_p$$

are given by

$$c_j = \frac{\mathbf{y} \cdot \mathbf{u}_j}{\mathbf{u}_j \cdot \mathbf{u}_j}, \; j = 1, ..., p.$$

Example 3: Express the vector $\mathbf{y} = \begin{bmatrix} 6 \\ 1 \\ -8 \end{bmatrix}$ as a linear combination of the vectors in S of Example 2, which is an orthogonal basis for $\mathbb{R}^3$.

Solution: Let us compute $\mathbf{y} \cdot \mathbf{u}_1 = 11$, $\mathbf{y} \cdot \mathbf{u}_2 = -12$, $\mathbf{y} \cdot \mathbf{u}_3 = -33$, $\mathbf{u}_1 \cdot \mathbf{u}_1 = 11$, $\mathbf{u}_2 \cdot \mathbf{u}_2 = 6$, $\mathbf{u}_3 \cdot \mathbf{u}_3 = 33/2$.

As $S = \{\mathbf{u}_1, \mathbf{u}_2, \mathbf{u}_3\}$ is an orthogonal basis (proved in Example 2), by the above theorem,

$$\begin{aligned} \mathbf{y} &= \frac{\mathbf{y} \cdot \mathbf{u}_1}{\mathbf{u}_1 \cdot \mathbf{u}_1}\mathbf{u}_1 + \frac{\mathbf{y} \cdot \mathbf{u}_2}{\mathbf{u}_2 \cdot \mathbf{u}_2}\mathbf{u}_2 + \frac{\mathbf{y} \cdot \mathbf{u}_3}{\mathbf{u}_3 \cdot \mathbf{u}_3}\mathbf{u}_3 \\ &= \frac{11}{11}\mathbf{u}_1 + \frac{-12}{6}\mathbf{u}_2 + \frac{-33}{33/2}\mathbf{u}_3 \\ &= \mathbf{u}_1 - 2\mathbf{u}_2 - 2\mathbf{u}_3 \end{aligned}$$

Definition 6.5. ***Orthonormal sets:*** *A set $\{\boldsymbol{u}_1, \cdots, \boldsymbol{u}_p\}$ is an* ***orthonormal set*** *if it is an orthogonal set of unit vectors. If W is the subspace spanned by such a set, then $\{\boldsymbol{u}_1, \cdots, \boldsymbol{u}_p\}$ is an orthonormal basis for W.*

Example 4: The simplest example of an orthonormal set is the standard basis $\{\mathbf{e}_1, \cdots, \mathbf{e}_n\}$ for $\mathbb{R}^n$. Any nonempty subset of $\{\mathbf{e}_1, \cdots, \mathbf{e}_n\}$ is orthonormal too.

Example 5: Show that $\{\mathbf{v}_1, \mathbf{v}_2, \mathbf{v}_3\}$ is an orthonormal basis of $\mathbb{R}^3$ where

$$\mathbf{v}_1 = \begin{bmatrix} 3/\sqrt{11} \\ 1/\sqrt{11} \\ 1/\sqrt{11} \end{bmatrix}, \ \mathbf{v}_2 = \begin{bmatrix} -1/\sqrt{6} \\ 2/\sqrt{6} \\ 1/\sqrt{6} \end{bmatrix}, \ \mathbf{v}_3 = \begin{bmatrix} -1/\sqrt{66} \\ -4/\sqrt{66} \\ 7/\sqrt{66} \end{bmatrix}$$

Solution:

$$\mathbf{v}_1 \cdot \mathbf{v}_2 = -3/\sqrt{66} + 2/\sqrt{66} + 1/\sqrt{66} = 0$$

$$\mathbf{v}_1 \cdot \mathbf{v}_3 = -3/\sqrt{726} - 4/\sqrt{726} + 7/\sqrt{726} = 0$$

$$\mathbf{v}_2 \cdot \mathbf{v}_3 = 1/\sqrt{396} - 8/\sqrt{396} + 7/\sqrt{396} = 0$$

Thus $\{\mathbf{v}_1, \mathbf{v}_2, \mathbf{v}_3\}$ is an orthogonal set. Also,

$$\mathbf{v}_1 \cdot \mathbf{v}_1 = 9/11 + 1/11 + 1/11 = 1$$

$$\mathbf{v}_2 \cdot \mathbf{v}_2 = 1/6 + 4/6 + 1/6 = 1$$

$$\mathbf{v}_3 \cdot \mathbf{v}_3 = 1/66 + 16/66 + 49/66 = 1$$

which shows that $\mathbf{v}_1$, $\mathbf{v}_2$, and $\mathbf{v}_3$ are unit vectors. Thus $\{\mathbf{v}_1, \mathbf{v}_2, \mathbf{v}_3\}$ is an orthonormal set. Since the set is linearly independent, they form a basis for $\mathbb{R}^3$.

Theorem: An $m \times n$ matrix U has orthonormal columns iff $U^T U = I$.
Theorem: Let U be an $m \times n$ matrix with orthonormal columns and let $\mathbf{x}$ and $\mathbf{y}$ be in $\mathbb{R}^n$ then
a. $\| U\mathbf{x} \| = \| \mathbf{x} \|$
b. $(U\mathbf{x}) \cdot (U\mathbf{y}) = \mathbf{x} \cdot \mathbf{y}$
c. $(U\mathbf{x}) \cdot (U\mathbf{y}) = 0$ iff $\mathbf{x} \cdot \mathbf{y} = 0$

Example 6: Let $U = \begin{bmatrix} 1/\sqrt{2} & 2/3 \\ 1/\sqrt{2} & -2/3 \\ 0 & 1/3 \end{bmatrix}$ and $\mathbf{x} = \begin{bmatrix} \sqrt{2} \\ 3 \end{bmatrix}$.
i) Does U have orthonormal columns?
ii) Verify that $\| U\mathbf{x} \| = \| \mathbf{x} \|$.
Solution: i) As $U^T U = I$, U has orthonormal columns.
ii) $U\mathbf{x} = \begin{bmatrix} 3 \\ -1 \\ 1 \end{bmatrix}$, $\| U\mathbf{x} \| = \sqrt{11}$, $\| \mathbf{x} \| = \sqrt{11}$. $\therefore \| U\mathbf{x} \| = \| \mathbf{x} \|$ is verified.

Assignment

1. Determine which sets of vectors are orthogonal.
a) $\begin{bmatrix} -1 \\ 4 \\ -3 \end{bmatrix}, \begin{bmatrix} 5 \\ 2 \\ 1 \end{bmatrix}, \begin{bmatrix} 3 \\ -4 \\ -7 \end{bmatrix}$, b) $\begin{bmatrix} 2 \\ -7 \\ -1 \end{bmatrix}, \begin{bmatrix} -6 \\ -3 \\ 9 \end{bmatrix}, \begin{bmatrix} 3 \\ 1 \\ -1 \end{bmatrix}$,

c) $\begin{bmatrix} 2 \\ -5 \\ -3 \end{bmatrix}, \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix}, \begin{bmatrix} 4 \\ -2 \\ 6 \end{bmatrix}$, d) $\begin{bmatrix} 3 \\ -2 \\ 1 \\ 3 \end{bmatrix}, \begin{bmatrix} -1 \\ 3 \\ -3 \\ 4 \end{bmatrix}, \begin{bmatrix} 3 \\ 8 \\ 7 \\ 0 \end{bmatrix}$

2. Show that $\{\mathbf{u}_1, \mathbf{u}_2\}$ is an orthogonal basis for $\mathbb{R}^2$. Then express $\mathbf{x} = \begin{bmatrix} 9 \\ -7 \end{bmatrix}$ as a linear combination of the vectors $\mathbf{u}_1 = \begin{bmatrix} 2 \\ -3 \end{bmatrix}$ and $\mathbf{u}_2 = \begin{bmatrix} 6 \\ 4 \end{bmatrix}$.

3. Show that $\{\mathbf{u}_1, \mathbf{u}_2, \mathbf{u}_3\}$ is an orthogonal basis for $\mathbb{R}^3$. Then express $\mathbf{x} = \begin{bmatrix} 8 \\ -4 \\ -3 \end{bmatrix}$ as a linear combination of the vectors $\mathbf{u}_1 = \begin{bmatrix} 1 \\ 0 \\ 1 \end{bmatrix}$, $\mathbf{u}_2 = \begin{bmatrix} -1 \\ 4 \\ 1 \end{bmatrix}$, and $\mathbf{u}_3 = \begin{bmatrix} 2 \\ 1 \\ -2 \end{bmatrix}$.

4. Determine which sets of vectors are orthonormal.

a) $\begin{bmatrix} 1/3 \\ 1/3 \\ 1/3 \end{bmatrix}, \begin{bmatrix} -1/2 \\ 0 \\ 1/2 \end{bmatrix}$, b) $\begin{bmatrix} -6 \\ 8 \end{bmatrix}, \begin{bmatrix} 8 \\ 6 \end{bmatrix}$,

c) $\begin{bmatrix} -2/3 \\ 1/3 \\ 2/3 \end{bmatrix}, \begin{bmatrix} 1/3 \\ 2/3 \\ 0 \end{bmatrix}$, d) $\begin{bmatrix} 1/\sqrt{10} \\ 3/\sqrt{20} \\ 3/\sqrt{20} \end{bmatrix}, \begin{bmatrix} 3/\sqrt{10} \\ -1/\sqrt{20} \\ -1/\sqrt{20} \end{bmatrix}, \begin{bmatrix} 0 \\ -1/\sqrt{2} \\ 1/\sqrt{2} \end{bmatrix}$

7. ORTHOGONAL PROJECTION

Definition 7.1. ***Orthogonal projection:*** *Given a nonzero vector* $\boldsymbol{u}$ *in* $\mathbb{R}^n$*, let us decompose a vector* $\boldsymbol{y}$ *in* $\mathbb{R}^n$ *into the sum of two vectors, one a multiple of* $\boldsymbol{u}$ *and the other orthogonal to* $\boldsymbol{u}$*. Thus,*

$$\mathbf{y} = \hat{\mathbf{y}} + \mathbf{z} \tag{7.1}$$

where $\hat{\mathbf{y}} = \alpha\mathbf{u}$ *for some scalar* α *and* $\boldsymbol{z}$ *is some vector orthogonal to* $\mathbf{u}$*. Given any scalar* α*, let* $\mathbf{z} = \mathbf{y} - \alpha\mathbf{u}$*, so that* (7.1) *is satisfied. Then* $\mathbf{y} - \hat{\mathbf{y}}$ *is orthogonal to* $\mathbf{u}$ *iff*

$$(\mathbf{y} - \alpha\mathbf{u}) \cdot \mathbf{u} = \mathbf{y} \cdot \mathbf{u} - (\alpha\mathbf{u}) \cdot \mathbf{u} = \mathbf{y} \cdot \mathbf{u} - \alpha(\mathbf{u} \cdot \mathbf{u}) = 0$$

i.e., (7.1) *is satisfied with* $\mathbf{z}$ *orthogonal to* $\mathbf{u}$ *iff* $\alpha = \frac{\mathbf{y}\cdot\mathbf{u}}{\mathbf{u}\cdot\mathbf{u}}$ *and* $\hat{\mathbf{y}} = \frac{\mathbf{y}\cdot\mathbf{u}}{\mathbf{u}\cdot\mathbf{u}}\mathbf{u}$*. The vector* $\hat{\mathbf{y}}$ *is called the* ***orthogonal projection of*** $\mathbf{y}$ ***onto*** $\mathbf{u}$ *and the vector* $\mathbf{z}$ *is called the* ***component of*** $\mathbf{y}$ ***orthogonal to*** $\mathbf{u}$*.*

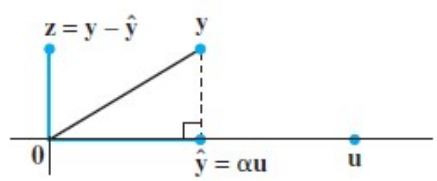


FIGURE 2
Finding α to make $\mathbf{y} - \hat{\mathbf{y}}$ orthogonal to $\mathbf{u}$.

Note: If c is any non-zero scalar and if $\mathbf{u}$ is replaced by $c\mathbf{u}$ in the definition of $\hat{\mathbf{y}}$, then the orthogonal projection of $\mathbf{y}$ onto $c\mathbf{u}$ is exactly the same as the orthogonal projection of $\mathbf{y}$ onto $\mathbf{u}$. Hence this projection is determined by the subspace L spanned by $\mathbf{u}$ (the line through $\mathbf{u}$ and $\mathbf{0}$). Sometimes $\hat{\mathbf{y}}$ is denoted by $\text{proj}_L\mathbf{y}$ and is called the **orthogonal projection of** $\mathbf{y}$ **onto** $\mathbf{L}$, i.e.,

$$\hat{y} = \text{proj}_L\mathbf{y} = \frac{\mathbf{y} \cdot \mathbf{u}}{\mathbf{u} \cdot \mathbf{u}}\mathbf{u} \tag{7.2}$$

Example 1: Let $\mathbf{y} = \begin{bmatrix} 7 \\ 6 \end{bmatrix}$ and $\mathbf{u} = \begin{bmatrix} 4 \\ 2 \end{bmatrix}$. Find the orthogonal projection of $\mathbf{y}$ onto $\mathbf{u}$. Then write $\mathbf{y}$ as the sum of two orthogonal vectors, one in $Span\,\{\mathbf{u}\}$ and one orthogonal to $\mathbf{u}$.

Solution: Let us compute

$$\mathbf{y} \cdot \mathbf{u} = \begin{bmatrix} 7 \\ 6 \end{bmatrix} \cdot \begin{bmatrix} 4 \\ 2 \end{bmatrix} = 40, \ \mathbf{u} \cdot \mathbf{u} = \begin{bmatrix} 4 \\ 2 \end{bmatrix} \cdot \begin{bmatrix} 4 \\ 2 \end{bmatrix} = 20$$

The orthogonal projection of $\mathbf{y}$ onto $\mathbf{u}$ is

$$\hat{\mathbf{y}} = \frac{\mathbf{y} \cdot \mathbf{u}}{\mathbf{u} \cdot \mathbf{u}}\mathbf{u} = \begin{bmatrix} 8 \\ 4 \end{bmatrix}$$

and the component of $\mathbf{y}$ orthogonal to $\mathbf{u}$ is

$$\mathbf{y} - \hat{\mathbf{y}} = \begin{bmatrix} -1 \\ 2 \end{bmatrix}$$

The sum of $\hat{\mathbf{y}}$ and $\mathbf{y} - \hat{\mathbf{y}}$ is $\mathbf{y}$, i.e.,

$$\begin{bmatrix} 7 \\ 6 \end{bmatrix} = \begin{bmatrix} 8 \\ 4 \end{bmatrix} + \begin{bmatrix} -1 \\ 2 \end{bmatrix}$$

(As a check, compute orthogonality i.e., $\hat{\mathbf{y}} \cdot (\mathbf{y} - \hat{\mathbf{y}}) = 0$)

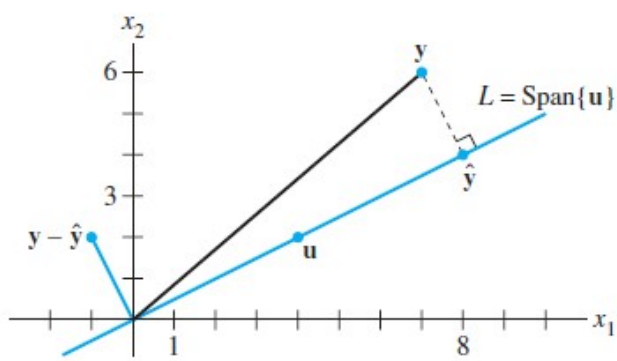


FIGURE 3 The orthogonal projection of $\mathbf{y}$ onto a line L through the origin.

Example 2: Find the distance in the above figure from $\mathbf{y}$ to L.

Solution: The distance from $\mathbf{y}$ to L is the length of the perpendicular line segment from $\mathbf{y}$ to the orthogonal projection $\hat{\mathbf{y}}$. This length equals the length of $\mathbf{y} - \hat{\mathbf{y}}$. Thus the distance is

$$\| \mathbf{y} - \hat{\mathbf{y}} \| = \sqrt{5}\,.$$

The orthogonal decomposition theorem: Let W be a subspace of $\mathbb{R}^n$.

Then each $\mathbf{y}$ in $\mathbb{R}^n$ can be written uniquely in the form

$$\mathbf{y} = \hat{\mathbf{y}} + \mathbf{z} \tag{7.3}$$

where $\hat{\mathbf{y}}$ is in W and $\mathbf{z}$ is in $W^{\perp}$. In fact, if $\{\mathbf{u}_1, \mathbf{u}_2, \cdots, \mathbf{u}_p\}$ is any orthogonal basis of W, then

$$\hat{\mathbf{y}} = \frac{\mathbf{y} \cdot \mathbf{u}_1}{\mathbf{u}_1 \cdot \mathbf{u}_1}\mathbf{u}_1 + \cdots + \frac{\mathbf{y} \cdot \mathbf{u}_p}{\mathbf{u}_p \cdot \mathbf{u}_p}\mathbf{u}_p \tag{7.4}$$

and $\mathbf{z} = \mathbf{y} - \hat{\mathbf{y}}$.

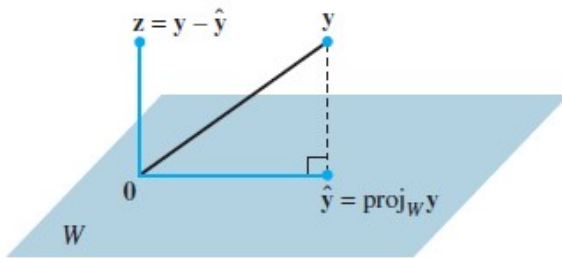


FIGURE 2 The orthogonal projection of $\mathbf{y}$ onto W.

The vector $\hat{\mathbf{y}}$ is called the orthogonal projection of $\mathbf{y}$ onto W and often written as $\text{proj}_W \mathbf{y}$.

Example 3: Let $\{\mathbf{u}_1, \cdots, \mathbf{u}_5\}$ be an orthogonal basis for $\mathbb{R}^5$ and let $\mathbf{y} = c_1\mathbf{u}_1 + \cdots + c_5\mathbf{u}_5$. Consider the subspace $W = Span\,\{\mathbf{u}_1, \mathbf{u}_2\}$ and write $\mathbf{y}$ as the sum of a vector $\mathbf{z}_1$ in W and a vector $\mathbf{z}_2$ in $W^{\perp}$.

Solution: Let us write $\mathbf{y} = \mathbf{z}_1 + \mathbf{z}_2$, where $\mathbf{z}_1 = c_1\mathbf{u}_1 + c_2\mathbf{u}_2$ is in Span$\{\mathbf{u}_1, \mathbf{u}_2\}$ and $\mathbf{z}_2 = c_3\mathbf{u}_3 + c_4\mathbf{u}_4 + c_5\mathbf{u}_5$ is in Span$\{\mathbf{u}_3, \mathbf{u}_4, \mathbf{u}_5\}$.

To show that $\mathbf{z}_2$ is in $W^{\perp}$, it suffices to show that $\mathbf{z}_2$ is orthogonal to the vectors in the basis $\{\mathbf{u}_1, \mathbf{u}_2\}$ for W. Let us compute

$$\mathbf{z}_2 \cdot \mathbf{u}_1 = (c_3\mathbf{u}_3 + c_4\mathbf{u}_4 + c_5\mathbf{u}_5) \cdot \mathbf{u}_1 = 0$$

because $\mathbf{u}_1$ is orthogonal to $\mathbf{u}_3$, $\mathbf{u}_4$, and $\mathbf{u}_5$. A similar calculation shows that $\mathbf{z}_2 \cdot \mathbf{u}_2 = 0$. Thus $\mathbf{z}_2$ is in $W^{\perp}$.

Example 4: Let $\mathbf{u}_1 = \begin{bmatrix} 2 \\ 5 \\ -1 \end{bmatrix}$, $\mathbf{u}_2 = \begin{bmatrix} -2 \\ 1 \\ 1 \end{bmatrix}$ and $\mathbf{y} = \begin{bmatrix} 1 \\ 2 \\ 3 \end{bmatrix}$. Observe that $\{\mathbf{u}_1, \mathbf{u}_2\}$ is an orthogonal basis for $W = Span\{\mathbf{u}_1, \mathbf{u}_2\}$. Write $\mathbf{y}$ as the sum of a vector in W and a vector orthogonal to W.

Solution: The orthogonal projection of $\mathbf{y}$ onto W is

$$\hat{\mathbf{y}} = \frac{\mathbf{y} \cdot \mathbf{u}_1}{\mathbf{u}_1 \cdot \mathbf{u}_1}\mathbf{u}_1 + \frac{\mathbf{y} \cdot \mathbf{u}_2}{\mathbf{u}_2 \cdot \mathbf{u}_2}\mathbf{u}_2 = \frac{9}{30}\begin{bmatrix} 2 \\ 5 \\ -1 \end{bmatrix} + \frac{3}{6}\begin{bmatrix} -2 \\ 1 \\ 1 \end{bmatrix} = \begin{bmatrix} -2/5 \\ 2 \\ 1/5 \end{bmatrix}$$

Also,

$$\mathbf{y} - \hat{\mathbf{y}} = \begin{bmatrix} 1 \\ 2 \\ 3 \end{bmatrix} - \begin{bmatrix} -2/5 \\ 2 \\ 1/5 \end{bmatrix} = \begin{bmatrix} 7/5 \\ 0 \\ 14/5 \end{bmatrix}$$

The orthogonal decomposition theorem ensures that $\mathbf{y} - \hat{\mathbf{y}}$ is in $W^{\perp}$. Let us verify that $\mathbf{y} - \hat{\mathbf{y}}$ is orthogonal to both $\mathbf{u}_1$ and $\mathbf{u}_2$ and hence to all of W. The desired decomposition of $\mathbf{y}$ is

$$\mathbf{y} = \begin{bmatrix} 1 \\ 2 \\ 3 \end{bmatrix} = \begin{bmatrix} -2/5 \\ 2 \\ 1/5 \end{bmatrix} + \begin{bmatrix} 7/5 \\ 0 \\ 14/5 \end{bmatrix}$$

Properties of orthogonal projections:

1. $\mathbf{y}$ is in $W = Span\{\mathbf{u}_1, \cdots, \mathbf{u}_p\}$, then $\text{proj}_W \mathbf{y} = \mathbf{y}$.
2. **The Best Approximation Theorem:** Let W be a subspace of $\mathbb{R}^n$, $\mathbf{y}$ be any vector in $\mathbb{R}^n$, and $\hat{\mathbf{y}}$ be the orthogonal projection of $\mathbf{y}$ onto W. Then $\hat{\mathbf{y}}$ is the closest point in W to $\mathbf{y}$ in the sense that $\| \mathbf{y} - \hat{\mathbf{y}} \| < \| \mathbf{y} - \mathbf{v} \|$ for all $\mathbf{v}$ in W distinct from $\hat{\mathbf{y}}$.
3. The distance from a point $\mathbf{y}$ in $\mathbb{R}^n$ to a subspace W is defined as the distance from $\mathbf{y}$ to the nearest point in W.

Example 5: If $\mathbf{u}_1 = \begin{bmatrix} 2 \\ 5 \\ -1 \end{bmatrix}$, $\mathbf{u}_2 = \begin{bmatrix} -2 \\ 1 \\ 1 \end{bmatrix}$, $\mathbf{y} = \begin{bmatrix} 1 \\ 2 \\ 3 \end{bmatrix}$ and $W = Span\{\mathbf{u}_1, \mathbf{u}_2\}$, as in Example 4, then the closest point in W to $\mathbf{y}$ is

$\hat{\mathbf{y}} = \frac{\mathbf{y}\cdot\mathbf{u}_1}{\mathbf{u}_1\cdot\mathbf{u}_1}\mathbf{u}_1 + \frac{\mathbf{y}\cdot\mathbf{u}_2}{\mathbf{u}_2\cdot\mathbf{u}_2}\mathbf{u}_2 = \begin{bmatrix} -2/5 \\ 2 \\ 1/5 \end{bmatrix}$.

Example 6: Find the distance from $\mathbf{y}$ to $W = Span\{\mathbf{u}_1, \mathbf{u}_2\}$, where $\mathbf{y} = \begin{bmatrix} -1 \\ -5 \\ 10 \end{bmatrix}$, $\mathbf{u}_1 = \begin{bmatrix} 5 \\ -2 \\ 1 \end{bmatrix}$, $\mathbf{u}_2 = \begin{bmatrix} 1 \\ 2 \\ -1 \end{bmatrix}$

Solution: By the Best Approximation Theorem, the distance from $\mathbf{y}$ to W is $\| \mathbf{y} - \hat{\mathbf{y}} \|$, where $\hat{\mathbf{y}} = \text{proj}_W\mathbf{y}$. Since $\{\mathbf{u}_1, \mathbf{u}_2\}$ is an orthogonal basis for W,

$\hat{\mathbf{y}} = \frac{15}{30}\mathbf{u}_1 + \frac{-21}{6}\mathbf{u}_2 = \begin{bmatrix} -1 \\ -8 \\ 4 \end{bmatrix}$

$\mathbf{y} - \hat{\mathbf{y}} = \begin{bmatrix} 0 \\ 3 \\ 6 \end{bmatrix}$, $\| \mathbf{y} - \hat{\mathbf{y}} \|^2 = 45$

$\therefore$ The distance from $\mathbf{y}$ to W is $\sqrt{45} = 3\sqrt{5}$.

8. GRAM-SCHMIDT ORTHOGONALIZATION PROCESS

Given a basis $\{\mathbf{x}_1, \cdots, \mathbf{x}_p\}$ for a nonzero subspace W of $\mathbb{R}^n$, define

$$\mathbf{v}_1 = \mathbf{x}_1$$

$$\mathbf{v}_2 = \mathbf{x}_2 - \frac{\mathbf{x}_2 \cdot \mathbf{v}_1}{\mathbf{v}_1 \cdot \mathbf{v}_1}\mathbf{v}_1$$

$$\mathbf{v}_3 = \mathbf{x}_3 - \frac{\mathbf{x}_3 \cdot \mathbf{v}_1}{\mathbf{v}_1 \cdot \mathbf{v}_1}\mathbf{v}_1 - \frac{\mathbf{x}_3 \cdot \mathbf{v}_2}{\mathbf{v}_2 \cdot \mathbf{v}_2}\mathbf{v}_2$$

$$\vdots \mathbf{v}_p = \mathbf{x}_p - \frac{\mathbf{x}_p \cdot \mathbf{v}_1}{\mathbf{v}_1 \cdot \mathbf{v}_1}\mathbf{v}_1 - \frac{\mathbf{x}_p \cdot \mathbf{v}_2}{\mathbf{v}_2 \cdot \mathbf{v}_2}\mathbf{v}_2 - \cdots - \frac{\mathbf{x}_p \cdot \mathbf{v}_{p-1}}{\mathbf{v}_{p-1} \cdot \mathbf{v}_{p-1}}\mathbf{v}_{p-1}$$

Then $\{\mathbf{v}_1, \cdots, \mathbf{v}_p\}$ is an orthogonal basis for W. In addition

$$Span\{\mathbf{v}_1, \cdots, \mathbf{v}_k\} = Span\{\mathbf{x}_1, \cdots, \mathbf{x}_k\} \text{ for } 1 \leq k \leq p.$$

Example 1: Let $W = Span\{\mathbf{x}_1, \mathbf{x}_2\}$, where $\mathbf{x}_1 = \begin{bmatrix} 3 \\ 6 \\ 0 \end{bmatrix}$ and $\mathbf{x}_2 = \begin{bmatrix} 1 \\ 2 \\ 2 \end{bmatrix}$.

Construct an orthogonal basis $\{\mathbf{v}_1, \mathbf{v}_2\}$ for W.

Solution: Let $\mathbf{v}_1 = \mathbf{x}_1$ and

$$\mathbf{v}_2 = \mathbf{x}_2 - \frac{\mathbf{x}_2 \cdot \mathbf{x}_1}{\mathbf{x}_1 \cdot \mathbf{x}_1}\mathbf{x}_1 = \begin{bmatrix} 1 \\ 2 \\ 2 \end{bmatrix} - \frac{15}{45}\begin{bmatrix} 3 \\ 6 \\ 0 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 2 \end{bmatrix}$$

Then $\{\mathbf{v}_1, \mathbf{v}_2\}$ is an orthogonal set of nonzero vectors in W. Since dim$W = 2$, the set $\{\mathbf{v}_1, \mathbf{v}_2\}$ is a basis for W.

Example 2: Let $\mathbf{x}_1 = \begin{bmatrix} 1 \\ 1 \\ 1 \\ 1 \end{bmatrix}$, $\mathbf{x}_2 = \begin{bmatrix} 0 \\ 1 \\ 1 \\ 1 \end{bmatrix}$, and $\mathbf{x}_3 = \begin{bmatrix} 0 \\ 0 \\ 1 \\ 1 \end{bmatrix}$. Then $\{\mathbf{x}_1, \mathbf{x}_2, \mathbf{x}_3\}$ is clearly linearly independent and thus is a basis for a subspace W of $\mathbb{R}^4$. Construct an orthogonal basis for W.

Solution: Step 1. Let $\mathbf{v}_1 = \mathbf{x}_1$ and $W_1 = Span\{\mathbf{x}_1\} = Span\{\mathbf{v}_1\}$.

Step 2. Let

$$\mathbf{v}_2 = \mathbf{x}_2 - \text{proj}_{W_1}\mathbf{x}_2 = \mathbf{x}_2 - \frac{\mathbf{x}_2 \cdot \mathbf{v}_1}{\mathbf{v}_1 \cdot \mathbf{v}_1}\mathbf{v}_1 \ (\because \ \mathbf{v}_1 = \mathbf{x}_1)$$

$$= \begin{bmatrix} 0 \\ 1 \\ 1 \\ 1 \end{bmatrix} - \frac{3}{4}\begin{bmatrix} 1 \\ 1 \\ 1 \\ 1 \end{bmatrix} = \begin{bmatrix} -3/4 \\ 1/4 \\ 1/4 \\ 1/4 \end{bmatrix}$$

$\mathbf{v}_2$ is the component of $\mathbf{x}_2$ orthogonal to $\mathbf{x}_1$, and $\{\mathbf{v}_1, \mathbf{v}_2\}$ is an orthogonal basis for the subspace W_2 spanned by $\mathbf{x}_1$ and $\mathbf{x}_2$.

Step 3.

$$\mathbf{v}_3 = \mathbf{x}_3 - \text{proj}_{W_2}\mathbf{x}_3 = \mathbf{x}_3 - \frac{\mathbf{x}_3 \cdot \mathbf{v}_1}{\mathbf{v}_1 \cdot \mathbf{v}_1}\mathbf{v}_1 - \frac{\mathbf{x}_3 \cdot \mathbf{v}_2}{\mathbf{v}_2 \cdot \mathbf{v}_2}\mathbf{v}_2$$

$$= \begin{bmatrix} 0 \\ 0 \\ 1 \\ 1 \end{bmatrix} - \frac{1}{2}\begin{bmatrix} 1 \\ 1 \\ 1 \\ 1 \end{bmatrix} - \frac{2}{3}\begin{bmatrix} -3/4 \\ 1/4 \\ 1/4 \\ 1/4 \end{bmatrix} = \begin{bmatrix} 0 \\ -2/3 \\ 1/3 \\ 1/3 \end{bmatrix}$$

Thus $\{\mathbf{v}_1, \mathbf{v}_2, \mathbf{v}_3\}$ is an orthogonal set of nonzero vectors and hence a linearly independent set in W. As W is three-dimensional ($\because$ it was defined by a basis of three vectors), $\{\mathbf{v}_1, \mathbf{v}_2, \mathbf{v}_3\}$ is an orthogonal basis for W.

Assignment

1. Compute the orthogonal projection of $\begin{bmatrix} 1 \\ -1 \end{bmatrix}$ onto the line through $\begin{bmatrix} -1 \\ 3 \end{bmatrix}$ and the origin.

2. Let $\mathbf{y} = \begin{bmatrix} 2 \\ 6 \end{bmatrix}$ and $\mathbf{u} = \begin{bmatrix} 7 \\ 1 \end{bmatrix}$. Write $\mathbf{y}$ as the sum of a vector in $Span\,\{\mathbf{u}\}$ and a vector orthogonal to $\mathbf{u}$.

3. Find the closest point to $\mathbf{y}$ in the subspace W spanned by $\mathbf{v}_1$ and $\mathbf{v}_2$, where $\mathbf{y} = \begin{bmatrix} 3 \\ 1 \\ 5 \\ 1 \end{bmatrix}$, $\mathbf{v}_1 = \begin{bmatrix} 3 \\ 1 \\ -1 \\ 1 \end{bmatrix}$, and $\mathbf{v}_2 = \begin{bmatrix} 1 \\ -1 \\ 1 \\ -1 \end{bmatrix}$.

4. $\left\{ \begin{bmatrix} 2 \\ -5 \\ 1 \end{bmatrix}, \begin{bmatrix} 4 \\ -1 \\ 2 \end{bmatrix} \right\}$ is a given set of basis for a subspace W. Use the Gram-Schmidt process to produce an orthogonal basis for W.

5. $\left\{ \begin{bmatrix} 0 \\ 4 \\ 2 \end{bmatrix}, \begin{bmatrix} 5 \\ 6 \\ -7 \end{bmatrix} \right\}$ is a given set of basis for a subspace W. Use the Gram-Schmidt process to produce an orthogonal basis for W.